

TEXT TO IMAGE

Adding your Hugging Face Token to Colab Secrets

1. **Get your HF Token:** Go to [Hugging Face settings](#) and generate a new access token (if you don't have one).
2. **Open Colab Secrets:** In your Colab notebook, click on the "🔑" (key) icon in the left sidebar.
3. **Add Secret:** Click 'Add new secret'.
4. **Name the Secret:** For the 'Name' field, type `HF_TOKEN` (this is case-sensitive and important).
5. **Paste your Token:** For the 'Value' field, paste your Hugging Face token.
6. **Save:** Ensure 'Notebook access' is enabled for this secret.

Once added, you can access the secret in your code like this:

```
!nvidia-smi
```

```
Thu Jul 2 17:28:57 2026
```

```
+-----+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----+-----+-----+-----+-----+-----+
| GPU  Name            Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap|      Memory-Usage | GPU-Util  Compute M. |
|                                           MIG M.           |
+-----+-----+-----+-----+-----+-----+
|   0   Tesla T4              Off   | 00000000:00:04:0 Off |                    0 |
| N/A   52C    P8             10W /  70W |  0MiB / 15360MiB |          0%      Default |
+-----+-----+-----+-----+-----+-----+
|
+-----+-----+-----+-----+-----+-----+
| Processes:
| GPU   GI    CI          PID    Type   Process name                      GPU Memory
|   ID   ID                                 Usage
+-----+-----+-----+-----+-----+-----+
| No running processes found
+-----+-----+-----+-----+-----+-----+
|
```

```
!pip install diffuser --upgrade
```

```
Collecting diffuser
  Downloading diffuser-0.0.1.tar.gz (1.6 kB)
  Preparing metadata (setup.py) ... done
Building wheels for collected packages: diffuser
  Building wheel for diffuser (setup.py) ... done
  Created wheel for diffuser: filename=diffuser-0.0.1-py3-none-any.whl size=1586 sha256=7ea8f9b8c25fc18a337d6a1bdb2161f86f74
  Stored in directory: /root/.cache/pip/wheels/9d/33/60/eaac3227d4a649668ab6c543d245f31202f68789c20df37aae
Successfully built diffuser
Installing collected packages: diffuser
Successfully installed diffuser-0.0.1
```

```
!pip install invisible_watermark transformers accelerate safetensors
```

```
Requirement already satisfied: PyWavelets>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark) (1.
Requirement already satisfied: numpy>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark) (2.0.2)
Requirement already satisfied: opencv-python>=4.1.0.25 in /usr/local/lib/python3.12/dist-packages (from invisible_watermar
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (from invisible_watermark) (2.11.0+cu128)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch->invisible_watermark)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch->invisible_watermark)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch->invisible_watermark)
```

```
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch->invisible_watermark)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-t
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolk
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-cuspars-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-tool
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[c
Requirement already satisfied: shellingham==1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1
Requirement already satisfied: rich==13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc==0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers)
Requirement already satisfied: cuda-pathfinder==1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: h11==0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: markdown-it-py==2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich==13.8.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich==13.8.0->type
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy==1.13.3->torch->i
Requirement already satisfied: MarkupSafe==2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch->invisible_w
Requirement already satisfied: mdurl==0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py==2.2.0->rich==13
Downloading invisible_watermark-0.2.0-py3-none-any.whl (1.6 MB)
1.6/1.6 MB 69.4 MB/s eta 0:00:00
Installing collected packages: invisible_watermark
```

```
import torch
from diffusers import DiffusionPipeline
import gradio as gr
# Load Stable Diffusion XL model
pipe = DiffusionPipeline.from_pretrained(
    "stabilityai/stable-diffusion-xl-base-1.0",
    torch_dtype=torch.float16,
    use_safetensors=True,
    variant="fp16"
)
pipe.to("cuda")
# Image generation function
def generate_image(prompt):
    image = pipe(prompt).images[0]
    return image
# Gradio Interface
demo = gr.Interface(
    fn=generate_image,
    inputs=gr.Textbox(label="Enter your prompt"),
    outputs=gr.Image(label="Generated Image"),
    title="Stable Diffusion XL Image Generator",
    description="Generate AI images using Stable Diffusion XL"
)
demo.launch()
```

Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your version of Flax.

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Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast downloads.

model_index.json: 100% 609/609 [00:00<00:00, 48.7kB/s]

Download complete: 100% 7.11G/7.11G [01:02<00:00, 132MB/s]

Fetching 19 files: 100% 19/19 [01:01<00:00, 3.84s/it]

Loading pipeline components...: 100% 7/7 [00:04<00:00, 1.75it/s]

Loading weights: 100% 196/196 [00:00<00:00, 393.63it/s]

Loading weights: 100% 517/517 [00:00<00:00, 692.09it/s]

It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share=True` for you.

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://baf411c3874e80a127.gradio.live>

This share link is temporary and will last for up to 1 week (best effort). For free permanent hosting and GPU upgrades, run `pip install gradio-cli` and `gradio cli`.

Stable Diffusion XL Image Generator

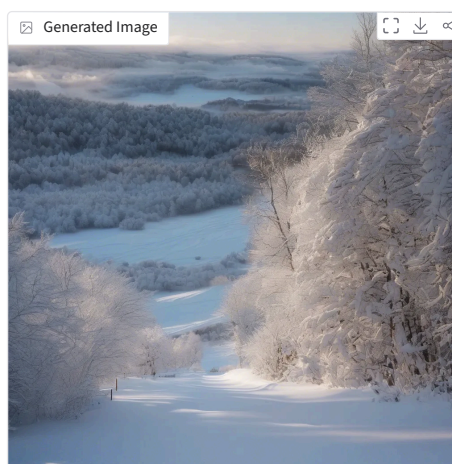
Generate AI images using Stable Diffusion XL

Enter your prompt

beautiful snow view

Clear

Submit



Flag